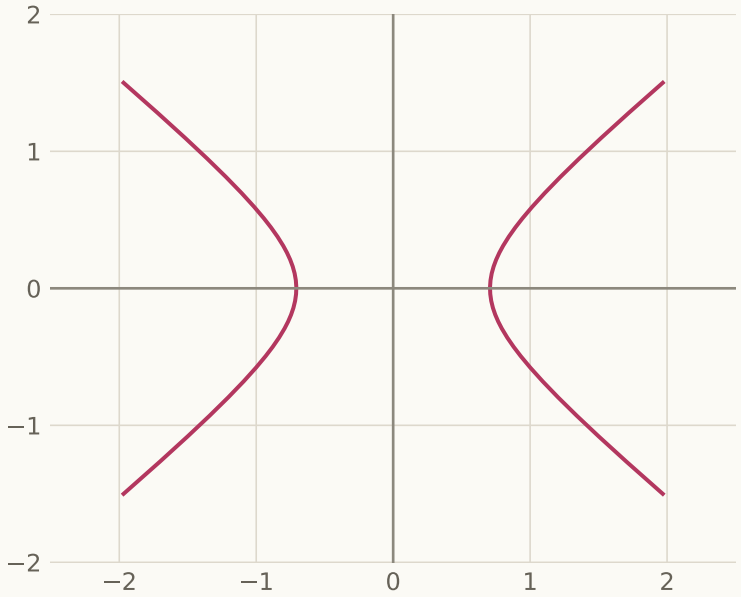
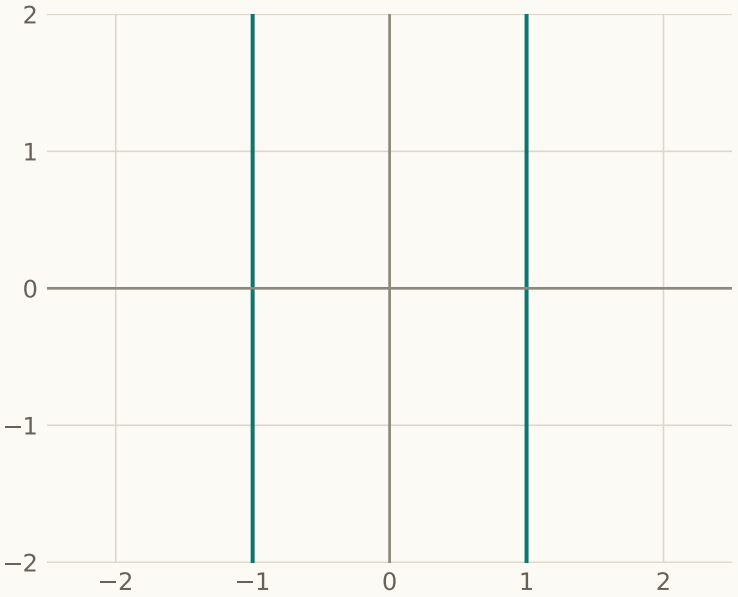
Mixed signs and zero directions change the level sets

indefinite: $2x^2 - 3y^2 = 1$



degenerate: $x^2 = 1$